

CCU004_03: Investigating the performance of the SCORE2 family-of-models during different phases of the COVID-19 pandemic.

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Version	Date	Authors	Comment
0.1	3/08/2023	Carmen Petitjean, Lisa Pennells, Spencer Keene, Angela Wood	Initial draft
0.2	26/09/2023	Carmen Petitjean, Lisa Pennells, Spencer Keene, Angela Wood	Comments from consortium members incorporated: Justification in text for not assessing QRISK in addition to SCORE2; clarification of exclusion criteria in text and Tables; justification of variable capture window in text; additional sensitivity analysis included – exclusion of individuals very recently hospitalized with COVID (in past 28 days); additional prior CVD history definitions added to exclusion criteria (Table 1b)

Background

Cardiovascular diseases (CVDs) are the leading cause of death globally, representing 1 in 3 global deaths (1). CVD deaths increased from 12.1 million in 1990 to 18.6 million in 2019 (2). Moreover, disability-adjusted life years (DALYs) and years of life lost also increased significantly globally with years lived with a disability doubling from 17.7 million to 34.4 million between 1990 and 2019 (2), which highlights the importance of early identification of high-risk individuals for recommendation of appropriate forms of effective primary prevention.

CVD risk prediction tools are primarily used to guide the uptake of prevention interventions, including behavioural (e.g., smoking cessation) and pharmaceutical interventions (e.g., statins) that aim to reduce the underlying CVD risk in the general population using a risk-stratification approach. These tools have the potential to also guide the intensity of interventions for CVD prevention for high-risk groups such as people with type 2 diabetes who are often already prescribed statins (3). The use of risk prediction tools is currently recommended in adults with no previous history of CVD every five years from 40 years of age as part of the NHS Health Check (4,5) and in guidelines from National Institute for Health and Care Excellence (NICE) (4) and the European Society of Cardiology (ESC) (6). CVD primary prevention guidelines for the UK (NICE) (4) and those from the ESC recommend the use of CVD risk prediction tools which consider demographic, lifestyle, physiological and biochemical parameters. NICE currently recommends the QRISK2 tool (7), whereas the ESC recommends the SCORE2 tool (8) to assess ten-year CVD risk (4). Model variations of SCORE2 exist, with SCORE2-OP (9) for individuals over 70 years of age and SCORE2-Diabetes (10) for individuals with type 2 diabetes.

Recent research findings from the BHF Data Science Centre suggest the COVID-19 pandemic has indirectly and directly induced a higher incidence of CVD. In particular, coronary heart disease and stroke incidences have been steadily rising since June 2020 and remain above the pre-pandemic rates.

The reasons for this are multiple. First, the use of medications to manage CVD was disrupted during the pandemic, resulting in thousands of individuals who are likely to have missed medication for CVD prevention and management (11). Second, the risk of CVD has been shown to be higher immediately following a COVID-19 diagnosis and remains elevated for at least 12 months (12). Elevated risks are highest among individuals who remain unvaccinated or under-vaccinated against COVID-19, estimated at around 30% of the adult population. Third, access to primary and secondary care has changed during the pandemic, resulting in lower primary care consultations per person and higher emergency hospital presentations.

An appealing design feature of the SCORE2 family-of-models is their integral recalibration system to tailor the models to different risk-regions (for low, moderate, high and very high-risk countries) across Europe based on registry data. Thus, the models can be readily updated to reflect recent CVD incidence and risk factor profiles of target populations. Hence, descriptive age- and sex-specific epidemiological data can be easily incorporated to rapidly revise models at a country or region level when needed. However, SCORE2 model performance has not been assessed in recent data collected since the COVID-19 pandemic and current versions of the models may under-estimate risk given the newly increased CVD rates.

In this study, we seek to investigate whether the SCORE2 family-of-models need recalibration to account for the changing CVD incidence (and possibly risk factor levels) due to the COVID-19 pandemic and, if so, to perform this recalibration for the population of England (and potentially Wales and Scotland). Furthermore, we aim to establish a pipeline that could rapidly update the model to account for future geographical, temporal, or demographical changes in CVD event rates.

Research Hypothesis

1. CVD risk estimated after the COVID-19 pandemic using current versions of SCORE2, SCORE2-OP and SCORE2-Diabetes underestimates observed CVD risk.
2. The degree of under-prediction may vary according to broad population characteristics, including geographical location, prior chronic condition status (including diabetes status), ethnicity, and the exposure to and severity of a COVID-19 infection.

Research questions

Using the SCORE-2, SCORE2-OP, and SCORE2-Diabetes risk estimation models we will address the following questions:

1. What is the performance (in terms of calibration and discrimination) of SCORE2, SCORE2-OP and SCORE2-Diabetes risk prediction models in their respective target populations after baseline risk assessment is performed at alternative time periods over the course of the COVID-19 pandemic:
 - a. Baseline risk assessment: 1st Jan 2020
 - b. Baseline risk assessment: 1st Jan 2022

2. Does the performance of SCORE2, SCORE2-OP and SCORE2-Diabetes vary according to the following subgroups of the population:
 - a. Those who have or have not been hospitalised with COVID-19
 - b. Those who have or have not been vaccinated against COVID-19
 - c. Individuals from different geographical regions
 - d. Individuals from different ethnic backgrounds
3. To what extent do the SCORE-2, SCORE2-OP and SCORE2-Diabetes risk prediction models need to be recalibrated to recent CVD incidence rates and risk factor values, for the whole population as well as recognising any identified subgroup differences in risk?

Data Sources

All data will be used up to the latest available stable release of data. We will first use the data from the National Health Service (NHS) Digital Secure Data Environment (SDE) for England, and later could possibly extend the analysis to the SAIL Databank for Wales and National Data Safe Haven for Scotland.

From the NHS Digital SDE:

- GPES Data for Pandemic Planning and Research (GDPPR) - Primary case data
- Secondary Use Service (SUS) Hospital data
- Hospital episode statistics Admitted Patient Care (HES APC)
- Office of National Statistics (ONS) death registration records
- COVID-19 SARI-Watch (formerly known as CHESS)
- Intensive Care National Audit and Research (ICNARC) COVID-19 data
- COVID-19 Second Generation Surveillance System (SGSS) - Pillar 1 and Pillar 2 COVID-19 infection laboratory testing data
- COVID-19 vaccination data
- NHS BSA Dispensed Medicines
- Secondary care medication (EPMA)

As these data were collected during the COVID-19 pandemic, inferences made might not be generalisable to pre- and post-pandemic time.

Study populations

Participants will be eligible to be included in the study if they meet the follow criteria:

- Between 40 and 89 years of age at baseline risk assessment
- Alive on January 1st 2020
- Have no previous history of CVD (Appendix)
- Have primary care (GDPPR) records available for research
- Live in England (or potentially Wales and Scotland if time permits)
- Have known sex and age

Follow-up

Start-dates (the date at which we will estimate risk) at the latest of:

- 1st of January 2020 (first cohort)
- 1st January 2022 (second cohort)

End of follow up will be at the earliest of:

- 12 months from the start date
- Date of death
- Date of non-fatal or fatal CVD event (see **Appendix** for CVD event definitions)

Exposures

We will consider SCORE2 and SCORE2-OP predictors (more detailed information is in the appendix with time windows below):

- Sex (binary)
- Age (in years, categorical)
- Systolic blood pressure (mmHg, continuous)
- Smoking status (current vs other, binary)
- Diabetes (type 2 diabetes, binary)
- Total cholesterol (mmol/L, continuous)
- HDL-cholesterol (mmol/L, continuous)

Furthermore, we will consider the additional SCORE2-Diabetes predictors among individuals with diabetes:

- Diabetes age of diagnosis (per 5-years, categorical)
- HbA1c (mmol/mol, continuous)
- eGFR (ml/min/1.73m², continuous)

Outcomes

Risk categories and CVD outcomes as defined in the development of the SCORE2 risk prediction models (see **Appendix**).

Additional and subgroup variables

We will consider the following additional variables:

- COVID-19:
 - COVID-19 infection status
 - Number of infection (first, second...)
 - Date of infection
 - Hospitalised for COVID-19
 - Critical care due to COVID-19
 - Date and type of COVID-19 vaccination
- Chronic conditions and date of diagnosis:

- Type 2 diabetes
- Chronic respiratory conditions including chronic obstructive pulmonary disease (COPD, interstitial lung disease (ILD), asthma and bronchiectasis.
- Ethnicity
- Sex
- Age groups
- Medication: statins, blood pressure medication and metformin
- Region of residence

Main analyses

First, we will compute initial descriptive statistics to characterise the baseline cohorts.

To ascertain CVD predictor values for SBP, HDL cholesterol, total cholesterol, HbA1c and eGFR, we will extract records between up to 18 months prior and 3-months after the start dates (i.e., date at which we will estimate each individual's CVD risk, in our case 1st January 2020 and 1st January 2022). This look-back and -forward period was selected to balance between the large proportions of missing data and for the second cohort to not be overly reliant on data measured in 2020 which is highly selective due to access to healthcare restrictions during the pandemic. To ensure the quality of selected measurements, we will define an acceptable range of values for continuous predictor values. To ascertain other CVD risk predictors such as smoking and diabetes and demographics such as ethnicity, we will extend the lookback period to any time prior to baseline. For more details on ascertainment timeframe *see Supplementary Table 2 in the **Appendix***. If multiple acceptable predictor values are recorded in the time-window, we will use the acceptable value recorded closest to baseline and will prioritise values recorded prior rather than after the start date. If no acceptable values were recorded in the time-window, the value of the predictor will be set as missing. Missing values will be later imputed (*see Missing Data section*). A similar approach was applied in previous work in the Clinical Practice Research data link, during derivation of SCORE2-Diabetes and validation of SCORE2. In this prior work, variable extraction up to 24 months prior and 6 months after baseline yielded sufficient data completeness. While we anticipate sufficient completeness in the current project, the time period for variable capture will be extended if necessary.

The CVD predictor values will be used to estimate 10-year CVD risk using the published SCORE2 models and coefficients on 1st January 2020 and on 1st January 2022. We will then assess the discrimination of the models by comparing the predicted risks with the observed risk over the 12 months of follow-up using Harrell's C-statistic. Due to the population-wide size of the dataset, we estimate that the cohorts would have enough events during the relatively short follow-up to achieve a stable discrimination estimate.

To assess calibration, average predicted risks will be compared to expected observed risks within sex and 5-year age groups. The expected (observed) 10-year risk for each sex and 5-year age groups will be estimated by extrapolating the 12-month observed CVD risk. Extrapolation will use the CVD rate at the mid-point of the coming 10-years, and assume this remains constant over the 10-year period. For example, the expected risk in 40–45-year-olds will be estimated using the annual CVD rate in the 45–49 year age group. Extrapolation will also take into account the competing risk of non-CVD death, using a life-table approach.

In addition to assessment of discrimination and calibration over the whole population, model performance will also be assessed within the following subgroups:

- By age groups as defined by ESC CVD risk threshold age categories (6): (40 to 49, 50 to 69 and 70+)
- Individuals who have and haven't been hospitalised due to COVID-19 infection, and individuals who have and haven't required critical care due to COVID-19 infection (as binary variables). This study aims to highlight how risk models can be amended to account for impact of the COVID-19 on CVD risk estimates. Focus on severe COVID-19 infections, rather the general community COVID-19 infection, will be able to consider its effect on individual's CVD risk, which may need to be separately recalibrated in the model or added as a risk predictor in the model. We expect the effect of the COVID-19 pandemic on the community would be broadly captured by recalibration to population-wide CVD event rates.
- Individuals who have and have not been fully vaccinated against COVID-19
- By ethnicity
- By region of residence

Should deviation from good calibration be found, either overall or in specific sub-groups of the population, recalibration of the models will be undertaken to improve predictive accuracy. Where appropriate the option of adding additional risk predictors to the model will be explored (e.g. addition of a predictor according to whether individuals have been previously hospitalised with COVID-19). The impact of including individuals hospitalised with COVID-19 within close proximity of the baseline date (e.g. within 28 days) will also be checked in sensitivity analysis, since the aim is to allow for long, rather than short term effects of severe COVID-19 on CVD risk.

Missing data

Individuals with missing age or sex will be excluded from this study.

For the missing values of the other predictors (i.e., SBP, HDL cholesterol, total cholesterol, smoking status for all SCORE2 models and HbA1c and eGFR for SCORE2-Diabetes), we will conduct multiple imputation by chained equations (MICE), by constructing imputation models including the following information: (13):

- CVD outcomes (time-to-event and censoring indicator - converted into the Nelson-Aalen estimator)

- Non-CVD deaths (time-to-event and censoring indicator - converted into the Nelson-Aalen estimator)
- all SCORE2/SCORE2-Diabetes predictors
- Postcode as a measure of deprivation
- Ethnicity
- Diagnosis of hypertension and hypotension
- Diagnosis of being overweight and underweight
- Number of GP visits in the last five years
- Number of hospital admission in the last five year

Due to the size of the dataset, imputation models will be derived on a 10% random subsample and then executed on the rest of the population. We will investigate the number of imputations needed and explore how to efficiently store the imputations. We expect it will be necessary to impute once, do the analysis, store the results, and repeat multiple times. Results from each imputation will then be combined using Rubin's rules for multiple imputation.

Future Extension

The established pipeline should also be readily applicable to QRISK models, which are currently recommended for use in England. However, assessment and recalibration of QRISK is beyond the scope and tight time frame of the current project since its more extensive list of predictors creates substantial complexity over the relatively simple SCORE2 models. Future work will be needed to accurately curate and impute the extensive list of additional QRISK variables needed for risk estimation as well as the corresponding population sex- and age group-specific prevalence that needs to be accurately estimated for recalibration. Careful incorporation the QRISK CVD endpoint will also be required since this is specified differently to that of the SCORE2 models and includes 'softer' less consistently recorded components such as Angina and Transient Ischemic Attack (TIA).

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Appendix

Supplementary table 1a: Outcomes considered by SCORE-2, table from the supplementary material (8)

Fatal cardiovascular disease– cause specific mortality due to any of the following:		
<i>Endpoints included</i>	<i>ICD10-codes</i>	<i>ICD9-codes</i>
Hypertensive disease	I10-16	401 – 405
Ischemic heart disease	I20-25	410 - 414
Arrhythmias, heart failure	I46-52	426 - 429
Cerebrovascular disease	I60-69	430 - 438
Atherosclerosis/AAA	I70-73	440 - 443
Sudden death and death within 24h of symptom onset	R96.0-96.1	798.1 , 798.2
 Endpoints excluded from the above endpoint:		
Myocarditis, unspecified	I51.4	426.7
Subarachnoid haemorrhage	I60	429
Subdural haemorrhage	I62	430
Cerebral aneurysm	I67.1	432.1
Cerebral arteritis	I68.2	437.3
Moyamoya	I67.5	437.4
 Non-fatal cardiovascular disease		
Non-fatal myocardial infarction	I21-I23	410
Non-fatal stroke	I60-69	430-438
 <i>Excluded from the non-fatal stroke endpoint:</i>		
Subarachnoid hemorrhage	I60	429
Subdural hemorrhage	I62	430
Cerebral aneurysm	I67.1	432.1
Cerebral arteritis	I68.2	437.3
Moyamoya	I67.5	437.4
 Death due to COVID-19		
Death from any cause other than CVD (defined above) or COVID-19		

Supplementary Table 1b Definition of history of vascular disease at baseline (used in exclusion criteria)

Prior disease	ICD code	
Coronary heart disease	I20-I25	410 - 414
Stroke	I60-69	430-438
TIA	G45	435.9
Peripheral artery disease	I70-71	440.2

Supplementary table 2: Detailed description of variables needed.

Variables	Notes	Timeframe	Source	
Co-variables (predictors)				
Systolic blood pressure (mmHg)	Not available prior to 2018	12 months prior and 3-month after prediction points – if multiple measurements are available, take the one closest to baseline. However, if one is before baseline always take that one.	Primary care records (GDPPR)	
Total cholesterol (mmol/L)	Not available prior to 2018			
HDL-cholesterol (mmol/L)	Not available prior to 2018			
Diabetes age of diagnosis				
HbA1c (mmol/mol)				
eGFR (ml/min/1.73m ²)				
Smoking status		Any time prior and 3-month after prediction points	Primary care records (GDPPR)	
Diabetes		Prior to the prediction points	Primary (GDPPR) and HES data	
Chronic conditions				
Diabetes type 2		Prior to the prediction points	Primary (GDPPR) and HES data	
Age at diabetes diagnosis	Use first diagnosis date			
COPD				
ILD				
Asthma				
Bronchiectasis				
COVID-19 variables				
COVID-19 infection status		Any time prior to prediction points and in the follow-up period	- COVID-19 SARI-Watch (formerly known as CHES) - COVID-19 Second Generation Surveillance System (SGSS) - Pillar 1 and Pillar 2 COVID-19 infection laboratory testing data - Hospital records in primary position (Hospital episode statistics Admitted Patient Care (HES APC))	
Date of infection				
Number of infection (first, second, third ... infection)	This one can be inferred from the data			
Hospitalised for COVID-19	Hospitalised within 28 days of a COVID-19 infection	Within 28 days of a COVID-19 infection		
Outcome				

CVD outcome	ICD-9/10 codes in the Appendix – only in primary position	After baseline, within follow-up	Primary position records for - Hospital episode statistics Admitted Patient Care (HES APC) - Death records
CVD outcome date			
Death due to COVID-19			
Non-COVID, non-CVD deaths			

Extra variables

Statins	<i>Not available prior to 2018</i>	1.5 years before baseline	Prescribed medicine dataset (NHS BSA Dispensed Medicine) – <i>Data not available prior to 2018</i>
Blood pressure medication	As defined in CCU018_01 and CCU002_01		
Metformin			
Insulin	Record date of first prescription to help ascertain diabetes type	Anytime	NHS BSA Dispensed Medicine, GPPR, HES
Number of GP visits in the last five year		Five years prior to prediction points	GPPR
Number of hospital visits in the last five year			Hospital episode statistics Admitted Patient Care (HES APC)
LSOA (deprivation index)		Five years prior and 6-month after prediction points (if multiple – take closest to baseline)	GPPR
Region of residence		The nine regions of England after mapping from latest recorded LSOA.	GPPR
Ethnicity		Any time	GPPR + HES
BMI			
Overweight/ Underweight diagnosis			GPPR + HES + medication (if any - otherwise build that list)
Hypertension/ Hypotension diagnosis			Use diagnosis code and medication
Eligibility criteria			
Age		At prediction points	
CVD History	To remove all individuals with a CVD history (ICD codes in	Any time before prediction points	GPPR, HES (use codes in both primary and secondary positions)

	Supplementary Table 1b)		
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